

1st International Conference on Soft Computing Application in Smart Grid and Renewable Energy (SAGAR2027)

26th & 27th February, 2027

Department of Electrical & Computer Engineering

&

Department of Electrical Engineering

Ajay Binay Institute Of Technology (Autonomous), Cuttack



About SAGAR2027 :

The **1st International Conference on Soft Computing Application in Smart Grid and Renewable Energy (SAGAR2027)** will be held on **February 26-27, 2027**, at **ABIT(Autonomous)** in **hybrid mode**. The conference is jointly organized by the **Department of Electrical and Computer Engineering** and **Department of Electrical Engineering**.

SAGAR2027 focuses on the application of **soft computing techniques** which includes **artificial intelligence, machine learning, neural networks, fuzzy logic, evolutionary algorithms, optimization, data science, and analytics** that address challenges in **smart grids and renewable energy systems**.

The conference aims to bring together researchers, academicians, industry professionals, and students to discuss recent advances in intelligent technologies for **power systems, renewable energy integration, energy management, forecasting, control, optimization, and reliability enhancement**. It provides a platform for knowledge sharing, collaboration, and exploration of future trends in sustainable and smart energy systems.

About Ajay Binay Institute of Technology (ABIT) Autonomous college :

Ajay Binay Institute of Technology (ABIT) is approved by **AICTE**, affiliated with **BPUT**, and has been granted **Autonomous Status by UGC**. ABIT group operates an **Engineering College** and an **Industrial Training Institute** in Cuttack, Odisha.

The group was established in **1993** with the aim to improve the availability of **skilled technical manpower** for industries as well as to enhance reach of technical education to rural Odisha.

The Institute is approved by **AICTE, New Delhi**, affiliated to **Biju Patnaik University of Technology, Rourkela** and **accredited by NAAC**. ABIT has a balanced distribution of technical streams with equal emphasis on the **core sector** and **IT sector**.

Important Date :

- **Submission Open: July 20, 2026**
- **Submission Deadline: Nov 20, 2026**
- **Acceptance Notification: Dec 27, 2026**
- **Author Registration: Jan 1, 2027**
- **Camera Ready Paper Submission: Jan 15, 2027**
- **Conference dates: February 26-27, 2027**

*Note:

- **Registration is valid for UG/PG students and PhD scholar with valid Id card, Academician and Industry professionals.**
- **At least one author registration is mandatory for each accepted paper.**
- **Co-authors who wish to attend the conference in offline mode must register under the Attendee category.**

***The proceedings are expected to be indexed in major databases such as Scopus.**

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Venue :

Department of Electrical and Computer Engineering (ECE) and Department of Electrical Engineering, AJAY BINAY INSTITUTE OF TECHNOLOGY (AUTONOMOUS), Markat Nagar, Cuttack Development Authority, Sector - 1, Cuttack - 753014

SAGAR2027 Website



<https://sagar2027.vercel.app/>

For Paper Submission



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TECHNICAL TRACKS:

Track 1: Power Systems and Smart Grids

- Power generation, transmission, and distribution
- Power system operation and control
- Smart grid technologies and applications
- Renewable energy integration and grid integration challenges
- Grid resilience and reliability enhancement
- Energy storage systems and management
- Demand-side management and demand response strategies
- Smart Grid Planning and Operation

Track 2: AI & ML application and Cyber security

- AI-Based Intrusion Detection Systems (IDS)
- Deep Learning Applications
- Phishing and Spam Detection Using Machine Learning
- Malware Classification with Deep Learning
- Network Anomaly Detection Using AI
- Adversarial Attacks and Defenses in Machine Learning

Track 3: Application of optimization techniques

- power system optimization
- Vehicle Routing and Logistics Optimization
- Hyperparameter Optimization in Machine Learning
- Smart Grid Energy Management Optimization
- Production and Inventory Scheduling Optimization
- Resource Allocation in Cloud Computing Using Optimization Algorithms

Track 4: Power Electronics and Intelligent Control Systems

- Intelligent control of power converters
- Advanced inverter technologies
- Control strategies for renewable energy systems
- FACTS devices and smart grid applications
- Adaptive and robust control techniques
- Embedded systems for smart energy applications
- Automation in power systems
- Electric Vehicle and Charging station

Track 5: Emerging Technologies and Future Trends

- Internet of Things (IoT) and communication techniques
- Quantum computing applications in energy systems
- Green hydrogen technologies
- AI-driven carbon neutrality solutions
- Internet of Energy (IoE)
- Smart energy markets and trading
- Digital transformation in power systems
- Future challenges in renewable energy and smart grids

Paper Submission Link: <https://cmt3.research.microsoft.com/SAGAR2027>

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|--------------------------------|--------------------------------|----------------------------|
| • Barabati Stadium | • Barabati Fort | • Chilika Lake |
| • Silvercity Boat Club | • Odisha State Maritime Museum | • Lingaraj Temple |
| • Shree Jagannath Temple, Puri | • Netaji Birth Place Museum | • Dhauli santi stupa |
| • Konark Sun Temple | • Cuttack Chandi Temple | • Khandagiri and Udayagiri |
| | | • Puri Sea Beach |



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